

CS772 Project Proposal

Team:- The Last Epoch

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Seed papers chosen:-

1. **Make Me a BNN:** A Simple Strategy for Estimating Bayesian Uncertainty from Pre-trained Models

- This paper proposes ABNN, a simple and efficient strategy to transform a pre-trained DNN into a BNN (Bayesian Neural Network) with uncertainty estimation capabilities.
- The method involves retraining only the normalization layer parameters (γ) and (β) of a pre-trained DNN. Multiple ABNN copies are trained to improve reliability and generalization.
- ABNN applies the Bayesian interpretation on the latent space rather than the weights, which reduces the variance of the gradients, resulting in more stable backpropagation.
- ABNN achieves competitive performance in terms of accuracy and uncertainty estimation on image classification and semantic segmentation tasks, in both in- and out-of-distribution settings.

2. **Prioritized Training on Points that are learnable, Worth Learning, and Not Yet Learnt**

- This paper introduces RHO-LOSS, a theoretically grounded selection function for prioritized training that enables substantial speedups, especially on noisy and web-scraped data.
- RHO-LOSS prioritizes points that are not yet learnt, learnable, and worth learning (non-noisy and task relevant), while deprioritizing points that are redundant (low training loss), noisy (high irreducible loss), or less relevant to the holdout set (high irreducible loss).
- Experiments show that RHO-LOSS can accelerate the training of various target architectures, even with a smaller irreducible loss model, and is robust to target-IL architecture mismatches.

3. **Dream the Impossible: Outlier Imagination with Diffusion Models**

- This paper introduces DREAM-OOD, a new framework for generating photo-realistic outliers using diffusion models, trained only on in-distribution (ID) data and classes.
- DREAM-OOD works by learning compact visual representations for the ID data, conditioned on the textual latent space of the diffusion model, and then sampling new visual embeddings in the text-conditioned latent space, which are then decoded into images.
- The synthesized outliers can regularize the ID classifier to be less confident in the out-of-distribution (OOD) region.

- Experiments show that DREAM-OOD improves OOD detection performance and can also be used to automatically generate ID samples, improving the generalization performance of the ID task itself.
- The method circumvents the challenge of describing outliers in words by learning a text-conditioned latent space based on ID data and sampling outliers in the low-likelihood region.

Tentative plan of action:-

1. Learnable Point Selection with RHO-LOSS

- Begin by using the Reducible Holdout Loss Selection (RHO-LOSS) technique.
- The RHO-LOSS calculation (Algorithm 1) involves computing both the training loss and the irreducible holdout loss to identify the most valuable points for training.

2. DNN Training

- Use the data points selected by RHO-LOSS to train a Deep Neural Network (DNN).
- Train a classifier on the selected data above to produce image embeddings, conditioned on the token embeddings.
- Monitor the training loss and validation performance to ensure the DNN is converging appropriately.

3. Conversion to Bayesian Neural Network (BNN) with ABNN

- Once the DNN is trained, convert it into a Bayesian Neural Network (BNN) using the Adaptable Bayesian Neural Network (ABNN) approach.
- ABNN transforms the pre-trained DNN into a BNN with uncertainty estimation capabilities by attaching simple BNN adaptation layers to normalization layers and fine-tuning only limited parameters with a limited number of epochs.

4. Regularization with DREAM-OOD Generated Outliers

- Employ the DREAM-OOD framework to generate photo-realistic outlier samples.
- This step helps the model to be less confident in the out-of-distribution regions and improves overall robustness.

Expected Outcome:-

By integrating RHO-LOSS, ABNN, and DREAM-OOD, the model achieves more reliable uncertainty estimation, improved generalization, and enhanced robustness against out-of-distribution data, resulting in a highly efficient and trustworthy deep learning system for real-world deployment.